

ISOM5240

Individual Assignment

Storytelling Application using Hugging Face Pipelines

Objective:

The objective of this assignment is to assess your Python programming skills and your ability to use Hugging Face's Transformers pipeline to build a storytelling application. The application should process an image provided by the user and generate a narrative. Additionally, the application must be designed to be deployed on Streamlit Cloud. The application is to be used and played by 3-10-year-old kids.

Learning Outcomes:

1. Develop programming solutions to solve business problems (PILO: 2, 3, 5, 7)
2. Write computer programs with common programming practices (PILO: 2, 3, 5)
3. Identify and fix logical and runtime errors in computer programs (PILO: 2, 3, 5)

Assignment Requirements:

1. Develop a Python-based storytelling application utilizing Hugging Face's Transformers pipeline.
2. Image input: Users upload an image, which the application processes.
3. Story generation: The application generates a narrative (50-100 words) based on details extracted from the uploaded image.
4. Text-to-Speech Conversion: The generated story should be converted into an audio format for an engaging experience.
5. Deployment: Deploy the final application on Streamlit Cloud so that users can interact with it online.

Implementation Guidelines:

1. Image Processing & Captioning:
 - Use a pre-trained image captioning model from Hugging Face to generate a caption from the uploaded image.
 - Example: Salesforce/blip-image-captioning-base model can be used.
2. Story Generation:
 - Use a text-generation model to expand the caption into a full story.
3. Text-to-Speech Conversion:

- Utilize a TTS (text-to-speech) model from Hugging Face or Python's pyttsx3 or gTTS module to convert the generated text into speech.
4. Deploy on Streamlit Cloud:
- Use Streamlit to create an interactive web UI.
 - Deploy the application to Streamlit Cloud following their guidelines.

Assessment Criteria:

Criterion	Description
Functionality	The application correctly processes images, generates a story, and converts it to audio.
Code Quality	Code is well-structured and documented. Implement the solution using functions for modularity and readability. Apply programming common practices , including meaningful variable names, proper indentation, and code documentation.
Model Usage	Appropriate pre-trained models from Hugging Face are utilized effectively.
User Experience	The application has an interactive and user-friendly UI using Streamlit.
Deployment	The application is successfully deployed on Streamlit Cloud.

Submission Instructions:

- Submit the following files:
 - app.py: Source code with comments
 - requirements.txt: Dependencies
 - Other files: Other required files
- Provide the Streamlit Cloud URL where the deployed application can be accessed.
- Ensure that all functionalities are tested and documented.

Late Submission Policy:

- Submissions up to 10 minutes late will incur a 30% penalty.
- Submissions more than 1 hour late will not be accepted.